Project SILAW (Spatial Illumination and Light Analysis for Wildlife): Assessing the Impact of Night-Time Light Pollution on Coastal and Marine Ecosystems in the Philippines

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Abstract

With a particular focus on Marine Protected Areas (MPAs), this study investigates the effects of nighttime light pollution on Philippine coastal and marine ecosystems. The Philippines, a major hub for marine biodiversity, is under tremendous ecological strain from urbanization and related light pollution, which interferes with vital life processes for fish and corals. Utilizing satellite data from 2014 to 2023, this study measures light pollution using the VIIRS Day/Night Band and evaluates risk levels using OCHA Philippines' MPA databases. By clustering MPAs using K-means to find natural groups and utilizing both ARIMA and Prophet models to predict future risk trends, the study employs a unique risk score computation approach. The findings show a noticeable rise in light pollution and related hazards over time, with some clusters—like those close to metropolitan areas such as Cebu City—facing increased risks as a result of urban development. The results highlight the necessity of community-driven interventions, sustainable urban development, and focused conservation initiatives to reduce the negative effects of light pollution on MPAs and promote marine ecosystem resilience.

I. Introduction

Due to its unique geographical location at the heart of the Coral Triangle, the Philippines is home to diverse marine wildlife. With its vast coral reefs, seagrass beds, and mangrove forests, this area is known as the global center of marine biodiversity and is home to an incredible variety of species. Some key marine species that thrive in the Philippines are whale sharks (*Rhincodon typus*), dugongs (*Dugong dugon*), sea turtles, thresher sharks (*Alopias pelagicus*), pygmy seahorses (*Hippocampus bargibanti/denise*), and manta rays (*Mobula birostris*). (Seller, 2016; Wormald, 2019) Millions of people who rely on fishing and tourism for their livelihoods are supported by the great biodiversity of the Philippine maritime ecosystem. More than 60% of people who live along the shore depend on marine resources for both their economic and nutritional needs. (Voices for Biodiversity, 2010) Innumerable species depend on coral reefs for vital habitat, which supports ecological balance and biodiversity worldwide.

However, the Philippines' abundant underwater ecosystems are seriously threatened by overfishing, habitat destruction, pollution, and climate change. These issues are made more difficult by the increase in light pollution brought on by urbanization, which disturbs marine life's natural habits, especially during crucial life stages like migration and breeding. (Marine Wildlife Watch of the Philippines, n.d.) Rapid urbanization presents serious problems in places like the Philippines because it encroaches on ecologically sensitive and coastal areas. Growing cities frequently result in more artificial illumination, which worsens light pollution and endangers biodiversity in the area. Because of their vast coasts and abundant biodiversity, the Philippines' distinctive ecosystems are especially at risk. (University of the Philippines Diliman - College of Science, n.d.)

Light pollution disrupts marine ecosystems by interfering with natural processes, disturbing ecological balance, inducing physiological stress in marine animals, and triggering a range of broader ecological impacts. (Portala, 2023) It disrupts patterns in migration of marine organisms that rely on natural light cycles from the moon and stars. This upsets the food webs as some of these organisms, which includes zooplanktons, are important in nutrient cycling, and are a primary food source for larger marine animals. (Ganguly & Candolin, 2023) Corals, like zooplanktons, are also affected by artificial lighting from urbanization. Their spawning calendar is based on natural light cycles, and light pollution interferes with the gametogenesis phase. This results in failure in reproduction, which was evident in the controlled experiments of Ayalon et al. (2021). Their study discovered that corals exposed to artificial light have delayed or impaired gamete development. Over time, it could lead to significant decline in coral populations. Artificial lighting may cause fish species that depend on the dark for protection to become more active, increasing the risk of predation. Fish exposed to artificial light show changed activity levels and irregular circadian rhythms, which over time may cause stress and even harm to their DNA. (Marangoni et al., 2022)

Light pollution also changes the dynamics between predator and prey. As mentioned earlier from the research of Marangoni et al. (2022), fish activity is influenced by light. Hence, it is now easier for predators to spot prey, and prey may find it harder to seek shelter. This could lead to poor checks and balances in the marine ecosystem. The easier hunting conditions also lead to imbalances in the ecosystem, decline in less adaptable species, and affect overall ecosystem health. Additionally, marine organisms may be further endangered by stress, which can impair immune responses and increase vulnerability to illnesses. Corals and their symbiotic algae only perform photosynthesis and nutrient exchange under specific light conditions. Disturbing these processes can lead disruptions in symbiotic relationships, resulting to reduced growth and increased mortality of these important species. (Carr, 2021; Miller & Rice, 2023)

Broader consequences of marine light pollution also include the disturbance of carbon cycling. In marine environments, zooplankton's diel vertical migration is essential to the cycling of carbon. The natural processes that sequester carbon may be impacted by light pollution's alteration of these migrations, which could influence global climate control efforts. (Miller & Rice, 2023) The unique variety of Philippine waters, which includes many endemic species already at risk from habitat loss and overfishing, is in danger from the cumulative effects of light pollution. Maintaining biodiversity and making sure that maritime habitats are resilient to other man-made stressors require protecting these ecosystems against light pollution. (Ganguly & Candolin, 2023)

To preserve marine resources and ecosystems, certain areas of the ocean, seas, or estuaries are designated as Marine Protected Areas (MPAs). Protecting biodiversity, preserving ecosystem integrity, and promoting the sustainable use of maritime resources are the core objectives of MPAs.

In the Philippines, there are over 1,500 designated MPAs, covering about 9.7% of its coastal waters. The administration and protection levels of the MPAs in the Philippines vary from one another. They consist of multiple-use areas that permit specific activities like fishing and tourism, as well as no-take zones that forbid any kind of extraction. Additionally, community-managed MPAs are common, where local people actively participate in conservation initiatives. (Baticados et al., 2004) Although MPAs are meant to be shielded from light pollution and other types of pollution, challenges such as inadequate regulations, urban encroachment, and lack of awareness may still render them vulnerable to light pollution. Currently, there may not be enough regulations that specifically address light pollution or even recognize its effects on marine ecosystems. Policymakers may also not be fully aware of the harm that light pollution causes on protected marine species. As urbanization reach coastal areas, especially on tourist destinations, increased artificial lighting may spill over into MPAs.

II. Statement of the Problem

This study aims to address the growing problem of light pollution in Philippine coastal and metropolitan areas and study its influence on marine biodiversity through analyzing the light pollution in Marine Protected Areas (MPAs). Specifically, the objectives of this study are:

- To retrieve light pollution data for the Philippines from satellite imagery from previous years;
- To retrieve data about Marine Protected Areas (MPAs) in the Philippines, locate them, and identify the marine wildlife that thrive in those areas;
- To calculate the risk score on the MPAs for the previous years using the yearly mean radiance values and the wildlife associated with the MPAs;
- To cluster MPAs according to mean radiance values and wildlife associated with the MPAs;
- To analyze the yearly trend of risk scores;
- And to develop a model that would accurately predict future risk score for each cluster.

Currently, there are limited studies about how nighttime light pollution impacts the fragile ecological balance in these MPAs, despite increasing urbanization and artificial lighting. Although the effects of artificial light on marine life are becoming more well recognized, there is still much to learn, according to a recent review of the literature on marine light pollution. In this study, Miller & Rice (2023) states that there is a gap of thorough research on how light pollution impacts numerous marine species and entire ecosystems, except for a few case studies. Additionally, aquatic areas have been the subject of comparatively few studies on artificial light at night (ALAN), with most of the studies focusing on terrestrial ecosystems. (Ganguly & Candolin, 2023) The impact of nocturnal light pollution on Philippine Marine Protected Areas (MPAs), which are crucial for marine biodiversity, is investigated in this study. It clusters MPAs to identify those most at danger, especially species that depend on natural light, such as fish and corals, and uses satellite data to compute risk ratings based on light exposure. The results are intended to inform conservation tactics and urban planning to lessen the effects of light pollution while striking a balance between ecological preservation and urban expansion.

This study is limited only to MPAs in the Philippines and utilizes the satellite data provided by the Earth Observation Group of Payne Institute for Public Policy Colorado School of Mines (n.d.) through Google Earth Engine. The data on MPAs were provided by OCHA Philippines (2018). This dataset includes the declared MPAs in the country for the year 2013. The results may be limited by data resolution, the intricacy of light's impacts on marine life, and possible biases in ecological data, which might compromise the precision and generalizability of results.

III. Methodology

Light pollution satellite data was retrieved from the Google Earth Engine. The Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB) nighttime data, which is supplied by the Earth Observation Group of Payne Institute for Public Policy Colorado School of Mines (n.d.) is used to create the monthly average radiance composite images of the database. Light pollution data was gathered for the years 2014-2023, clipped within the boundaries of the Philippines, with a resolution of 250 m/pixel. The mean radiance values were aggregated yearly and measured in nanoWatts/sr/cm^2. Data on marine protected areas were also gathered from OCHA Philippines (2018) through the Humanitarian Data Exchange website. The MPA dataset contains 339 MPAs, along with their names, geographical locations, MPA types, and environmental features.

The MPA and the VIIRS dataset were checked for consistency through their Coordinate Reference Systems (CRS), bounds, and overlap. The maps were also overlayed to check for compatibility and consistency with each other. If necessary, maps were reprojected to a common CRS to match each other. MPAs that do not overlap the raster files were dropped altogether.

Once the datasets and the maps were verified to be consistent, light pollution analyses were conducted. Mean radiance values were extracted from the VIIRS for each MPA for each year, from which risk scores were derived. The formula for calculating risk scores is:

$$R = k_1\left(\frac{\bar{r}}{p}\right) + k_2 e$$

Where R = risk score

k_1 = weight of light pollution

$\bar{r}$ = mean average radiance, in nW/sr/cm^2

p = high light pollution threshold

k_2 = weight of environmental features

e = environmental score, calculated as:

$$e = \frac{(2M + 2L + 3C + 2S + 2G + \frac{AC}{1000}}{maximum\ possible\ score}$$

Where M = presence of marine life

L = presence of littoral

C = presence of corals

S = presence of seagrass

G = presence of mangroves

AC = coral reef area, in m^2

$maximum\ possible\ score = 2 + 2 + 3 + 2 + 2 + \frac{\max(AC,1)}{1000}$.

This formula accounts for the light pollution received by the area, measures it against the threshold of light pollution that are considered harmful to existing marine wildlife, and considers areas with more environmental features to be more vulnerable to light pollution. Although there are no sources that directly state how much light pollution is damaging to marine life, this study estimates it to be 0.3 $nW/sr/cm^2$. This is a reasonable estimate as research suggests that even low level light pollution can be detrimental to marine life, especially corals. (Ayalon et al., 2019; Dybas, 2020; Rosenberg et al., 2019) Then, risk levels were classified according to this scale:

- Low Risk: $R \leq 0.25$
- Medium Risk: $0.25 < R \leq 0.5$
- High Risk: $0.5 < R \leq 0.75$
- Very High Risk: $0.75 < R \leq 1$
- Critical Risk: $R > 1$

Quantified risk levels were used to create light pollution risk maps which shows which MPAs are in danger from light pollution. This generated insights on how urbanization has been affecting MPAs and the Philippines, and how the risk scores has changed over time.

The MPAs were afterwards grouped according to their natural grouping patterns using K-means clustering. This was done using the average light pollution they receive and their environmental characteristics. K-means clustering is an unsupervised machine learning algorithm that divides data into a predetermined number of clusters. K-means seeks to produce compact, discrete groups by minimizing the sum of squared distances inside clusters. The algorithm starts by randomly choosing k initial centroids, which serve as the central points of the clusters. After that, it creates clusters according to distance by allocating each data point to the closest centroid. Following this assignment, the average locations of every data point inside each cluster are used to compute the centroids. Iteratively, points are assigned, and centroids are updated until the centroids stabilize, or a certain number of iterations is reached. that differ in density and size, which makes it ideal for data having distinct, well-defined groups. These clusters were evaluated using Silhouette scores. By comparing how well each data point fits into its designated cluster to other clusters, silhouette scores are a statistic used to assess the quality of clustering findings. Better-defined and more cohesive clusters are indicated by a higher score, which runs from -1 to 1. While a score close to 0 indicates that a data point is on or near the boundary between clusters, a score close to 1 shows that a data point is poorly matched to surrounding clusters and well matched to its own cluster. A data point may have been allocated to the incorrect cluster if the value is negative. The best number of clusters for a dataset may be found with the use of silhouette scores, which offer an easy-to-understand method of evaluating the cohesiveness and separation of clusters.

Optimal K-means clusters were calculated using the Elbow Method. It is an approach for figuring out the ideal number of clusters in a clustering algorithm, such as K-means. Plotting the explained variance, also known as the within-cluster sum of squared distances (SSE), against the number of clusters allows one to determine the "elbow point" at which the sum of squared distances' rate of decline sharply decreases. Since adding additional clusters after this point results in declining rewards in terms of explained variance, this point reflects the ideal number of clusters. The elbow approach is frequently used in clustering applications to prevent overfitting or underfitting of the data since it offers a clear and simple means of striking a balance between model complexity and accuracy.

Future risk scores were calculated for all MPAs in the Philippines as well as per cluster level using Autoregressive Integrated Moving Average (ARIMA) and Facebook Prophet. ARIMA, is a popular statistical modeling method for time series forecasting. It consists of three parts: Moving Average (MA), which models the relationship between an observation and a residual error from a moving average model applied to previous observations; Autoregression (AR), which models the relationship between a variable and its past values; and Differencing (I), which makes the time series stationary by eliminating trends and seasonality. Three parameters define the ARIMA model: the degree of differencing (d), the order of the MA term (q), and the order of the AR term (p). ARIMA assists in producing precise future value forecasts for time-dependent data by identifying patterns including trends, cycles, and seasonality.

Meanwhile, Facebook created Prophet, an open-source forecasting tool, especially for time series data with significant patterns, outliers, and seasonality. Even with little data, it generates reliable forecasts by automatically handling missing data, trend shifts, and holiday impacts. Prophet may identify daily, weekly, and annual patterns by employing an additive model that integrates trend, seasonality, and holiday influences. Its interpretability and ease of usage make it a desirable option for forecasting activities in fields like marketing, sales, and other areas where comprehending trends and future forecasts is essential.

The prediction models were trained on the historical data from 2014 to 2022 and evaluated on the year 2023, using a simple error metric, the root mean squared error (RMSE) score. It is a common statistic for assessing how well a model predicts outcomes in proportion to observable data. By using the square root of the average of the squared differences, it measures the discrepancy between expected and actual values. Better performance and accuracy are shown by a lower RMSE number, which shows that the model's predictions are more in line with the real data. Because it penalizes large mistakes more severely because of the squaring of differences, RMSE is more sensitive to large errors and may be used to discover models that often generate significant prediction errors. RMSE is widely used in regression studies, such as time series forecasting and other predictive modeling applications. Finally, using the better of the two models, risk scores for the next 5 years (from 2023) were forecasted.

IV. Results and Discussion

A total of 339 records of MPAs contained in the MPA dataset from OCHA. Most of the MPAs are located around Cebu, Bohol and Negros islands. The VIIRS raster maps and the MPA shapefiles were set to a common CRS for proper plotting. The Philippines is divided into Universal Transverse Mercator (UTM) zones 50 to 51, so to solve that, the maps are projected to the Philippine Albers Equal Area projection (EPSG code 3123). It is a customized coordinate reference system (CRS) created especially for mapping the whole Philippines with the least amount of distortion. Because it is a modification of the Albers Equal Area Conic projection, which maintains area attributes, it is ideal for analyses like resource management, distribution studies, and land use that call for precise representations of areas. Unlike the UTM, which is a cylindrical projection, the Albers Equal Area is a conic projection. Verifications show that the MPA shapefile and the VIIRS raster files do have an overlap and are consistent with each other. However, upon further inspection, the shapefile contains only 199 valid geometries. The locations of these MPAs with valid geometries are shown in Figure 1.

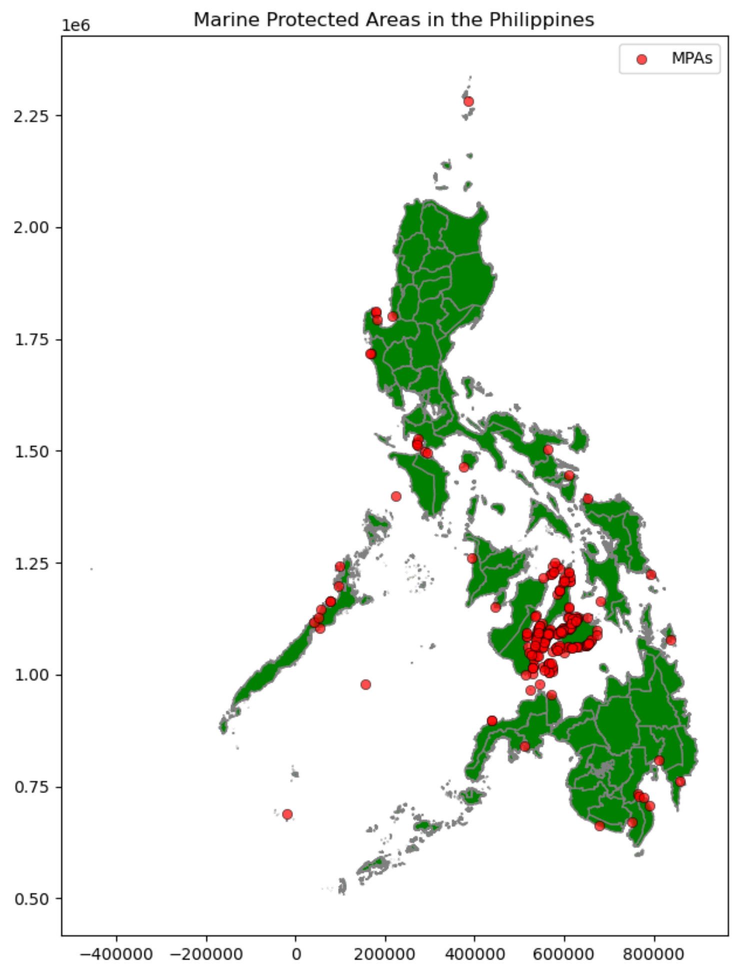

Figure 1. Locations of Marine Protected Areas in the Philippines.

Analysis of the mean radiance values for all MPAs show an overall increasing trend, shown in Figure 2. With notable variations possibly corresponding to particular events, policies, or socioeconomic factors impacting artificial lighting or human activities around MPAs, this trend points to a general increase in urbanization or human activity contributing to increased light pollution in MPAs over time. One particular event is the Covid-19 pandemic in 2020. It can be observed that the mean radiance in MPAs have been steadily increasing up to 2020, which points to increased human activity around coastlines, implying a general increase in tourism activities. However, when pandemic restrictions were implemented, the light pollution dipped, indicating a halt in human activity along the coastlines. In 2023, pandemic restrictions were lifted, which we can see in the graph as a sudden increase in light pollution values.

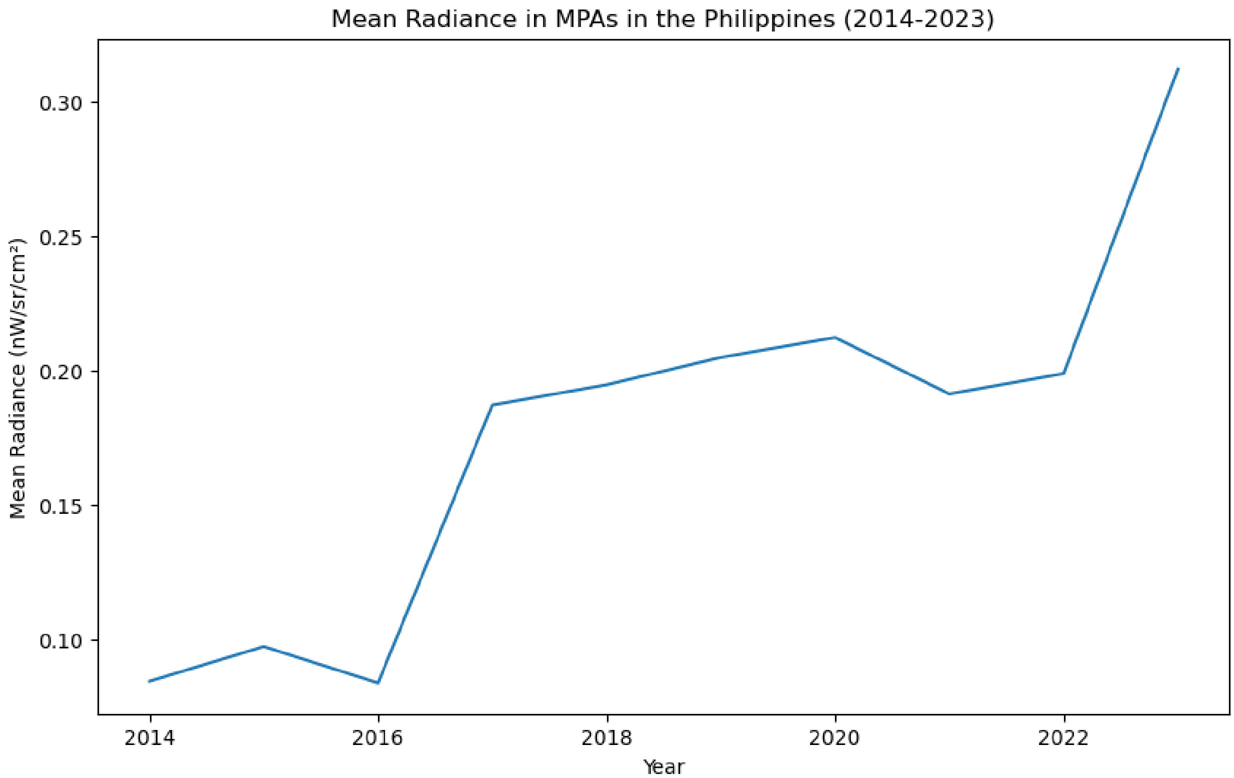

Figure 2. Light Pollution Values in MPAs from 2014-2023.

Calculation of risk scores using the provided formula gave insights as to how the MPAs are being affected by the increasing light pollution over the years. Since the risk scores are dependent on the light pollution values and the environmental factors, their trend is about the same as the trend of light pollution values. Figure 3 shows the progression of risk scores for all MPAs, with colored bands indicating low risk (green), medium risk (yellow-green), high risk (yellow), very high risk (orange), and critical risk (red). On average, the risk score of the MPAs of the Philippines are in critical condition as of 2023.

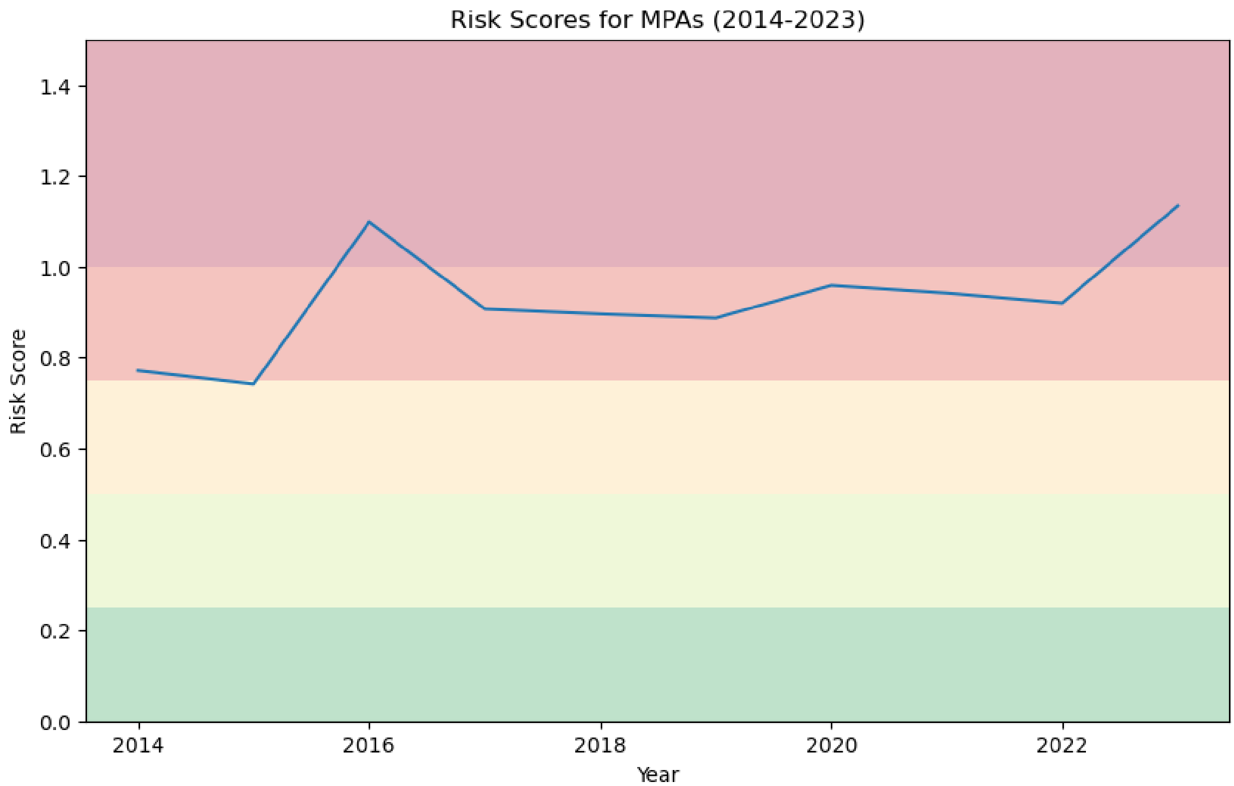

Figure 3. Risk Scores in MPAs, 2014-2023.

Figure 4 shows heat maps of MPAs, classified according to risk levels, from the year 2014 up to 2023. All the MPAs have progressed from a lower risk category to a higher risk category throughout the years, shown in the figure as the dots reddening as the years advance. Even those in remote areas like Batanes, Sulu, and islands between Zamboanga and Palawan have moved from low-risk category to critical risk category.

The average yearly change in the risk score from 2014 to 2023 for each MPA was documented and illustrated in Figure 5. This is calculated by calculating the total change in risk score divided by the number of years. Red and orange colors highlight areas where risk scores have increased the most on an annual basis, indicating a consistent rise in threats likely due to intensified human activities, such as urbanization, coastal development, or light pollution. Green and yellow colors indicate areas with a decrease or minimal change in risk scores, suggesting possible successful local conservation measures, reduced environmental stress, or effective mitigation efforts. The geographic distribution of these changes reveals that many coastal regions, especially those near urban centers, experience persistent increases in risk, emphasizing the impact of anthropogenic factors on marine ecosystems. Conversely, isolated areas with declining risk highlight the potential benefits of focused conservation strategies. This spatial pattern suggests a need for targeted, region-specific interventions and sustained efforts to manage human impacts and protect vulnerable marine ecosystems.

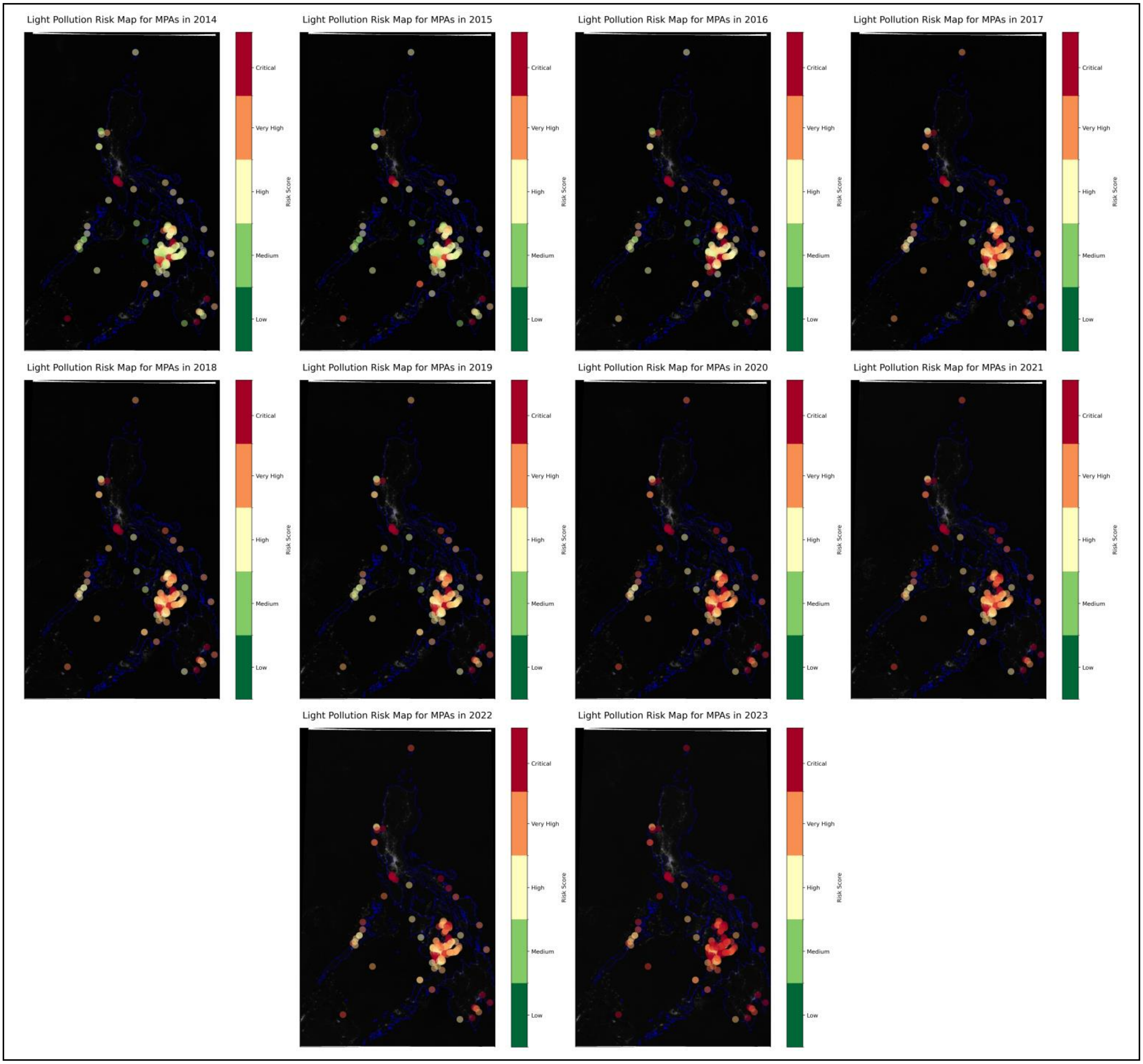

Figure 4. Heat Maps of Risk Scores per MPA from 2014 to 2023 (with Light Pollution Map Overlay).

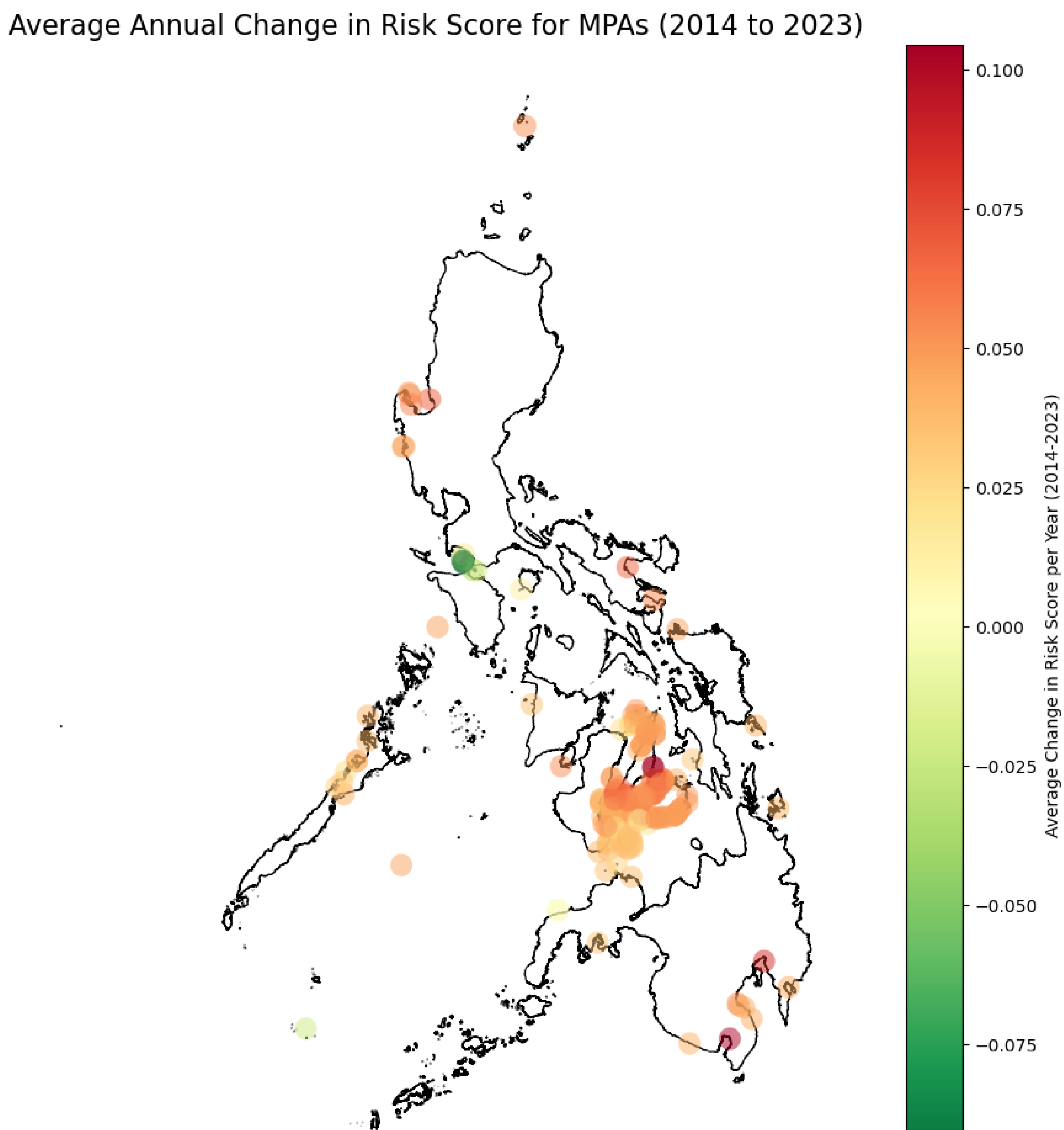

Figure 5. Changes in Risks Scores per MPA (2014-2023)

Using K-means clustering, the natural grouping among MPAs were identified using features such as average light pollution and environmental features. Environmental features in the dataset are existing marine life (MARINE_EXI), existing littorals (LITTORAL_E), existing corals (CORAL_EXIS), existing seagrass (SEAGRASS_E), existing mangroves (MANGROVE_E), existing corals (CORAL_EXIS), and reef area, in m^2 (REEF_AREA_). All MPAs had existing marine life and but none of them had littoral, so these features were removed from the clustering. The remaining features and their correlation with each other are shown in Figure 6. It is important that none of the features are highly correlated with another feature, so that distinct natural groupings among MPAs can be identified.

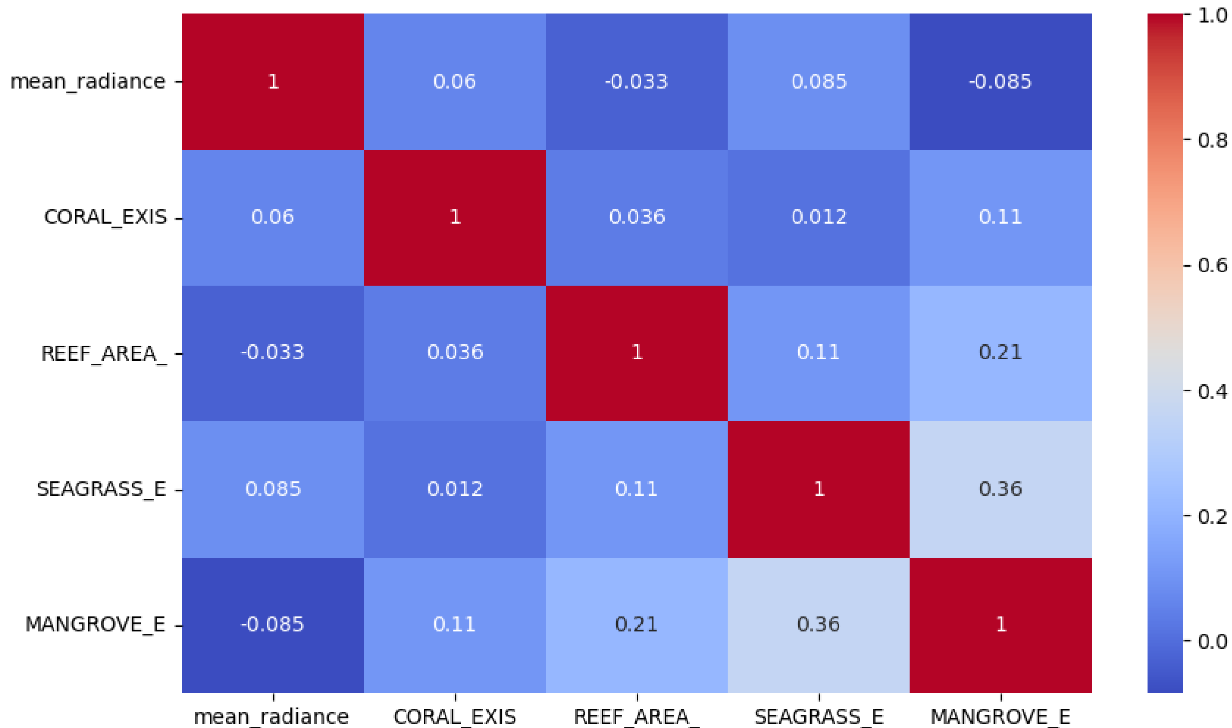

Figure 6. Correlation Matrix of Features Used for Clustering

The optimal number of clusters for the MPAs was calculated using the Elbow Method, shown in Figure 7. The elbow point appears to be around 4-6 clusters, which is where the greatest decrease in SSE can be observed. Beyond 6 clusters, there is only slight changes in SSE as the clusters increase. For this study, 6 clusters were selected as optimal, since this is where the elbow point is prominent. Upon model evaluation, the average silhouette score for all was calculated to be 0.7502. The silhouette plot for each cluster is in Figure 8. This score indicates that the points fit very well within each cluster while also fitting poorly with other nearby clusters, suggesting excellent clustering.

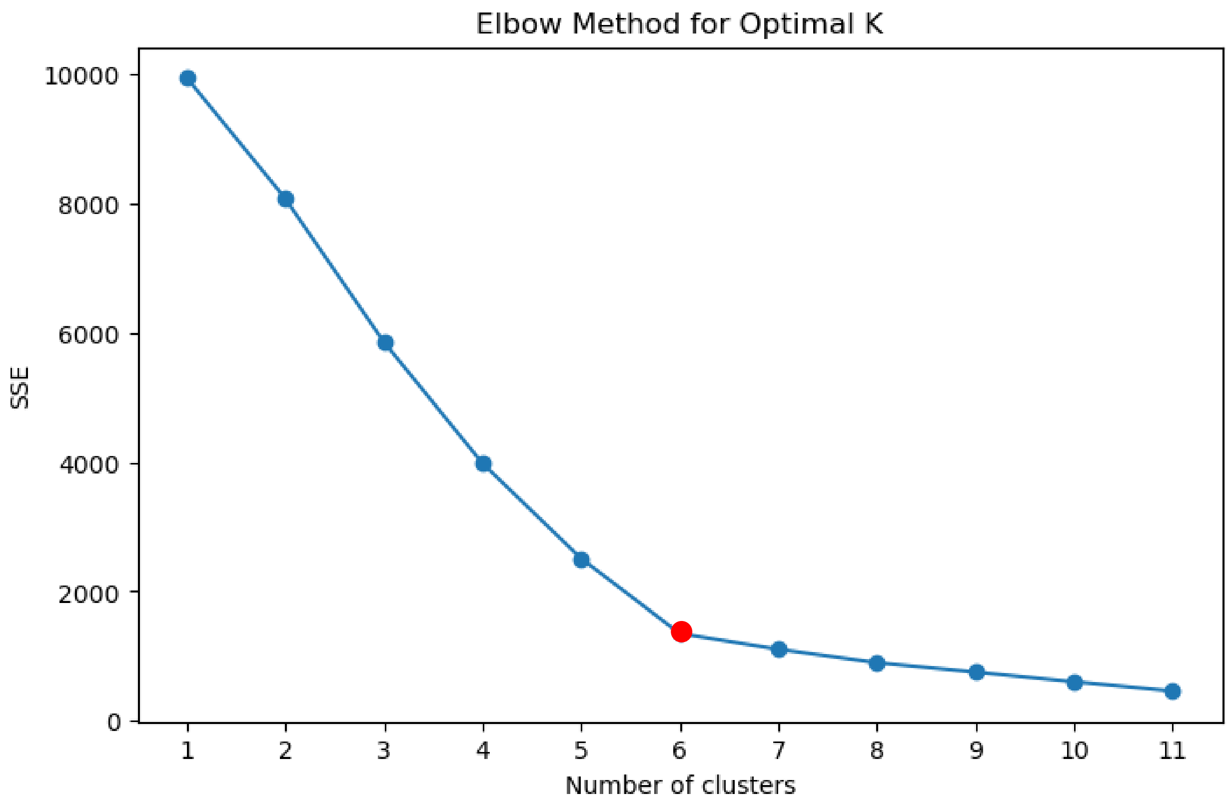

Figure 7. Elbow Method for Optimizing Clusters.

The common characteristics of MPAs within each cluster were analyzed in Figure 9. Cluster 5 MPAs receive the most light pollution on average, significantly higher than the rest of the clusters, while cluster 0 has the least light pollution. This indicates the proximity of cluster 5 MPAs to urban areas. In terms of marine biodiversity, cluster 0 and 2 has all the types of marine life indicated in the MPA database, while cluster 1 is the least diverse. Among the MPAs, cluster 0 has the widest area of coral reef coverage.

The natural grouping of the MPAs can be describe further as follows:

Cluster 0 - ***Expansive Coral-Dominated Ecosystem:*** This cluster features the largest coral area by a significant margin. It also shows a relatively balanced distribution of other environmental features like seagrass, mangroves, and marine ecosystems. This cluster represents expansive coral-dominated regions, likely with interconnected ecosystems. Its large coral area makes it a key conservation priority due to its ecological significance and potential vulnerability to disturbances.

Cluster 1 - ***Moderate Radiance Coastal Zone:*** This cluster shows moderate radiance levels, and a less diverse distribution of environmental features compared to other clusters. This clusters might represent coastal MPAs with moderate human impact and potentially less complex or diverse ecosystems. It may still experience significant human influence but does not reach the extremes seen in other clusters.

Cluster 2 - ***Diverse Coral-Seagrass System:*** This cluster exhibits a balanced mix of coral, seagrass, and mangrove features, with a notable amount of seagrass coverage. The diverse composition suggests a complex ecosystem with high ecological interdependence. This cluster may be sensitive to disturbances affecting any single feature and requires holistic conservation measures.

Cluster 3 - ***Coral-Enriched Area with Moderate Disturbance:*** This cluster has a moderate coral presence and a notable number of other features. It exhibits average radiance levels, indicating moderate human impact. This cluster represents regions where coral ecosystems are present but not dominant compared to Cluster 0. The human influence appears moderate, suggesting a mix of natural and anthropogenic factors shaping its risk profile.

Cluster 4 - ***Mangrove-Dominated Buffer Zone:*** This cluster shows a higher proportion of mangroves compared to other features and relatively moderate radiance levels. This cluster appears to serve as a natural buffer zone, potentially reducing the impact of human activity on nearby ecosystems. Conservation strategies here could focus on mangrove protection due to its ecological role in providing resilience to adjacent ecosystems.

Cluster 5 - ***Urban-Impacted High Radiance Zone:*** This cluster stands out due to its high mean radiance levels, indicating intense human activity or urban influence. The presence of other environmental features is less emphasized compared to radiance. This cluster likely represents MPAs heavily influenced by urbanization and high light pollution, making it a priority for managing anthropogenic impacts and mitigating light pollution.

The locations of these clusters are shown in Figure 10. It can be observed that cluster 5 MPAs, which can be seen as most affected by light pollution, are concentrated around the coastlines of Cebu City, a major urban and economic hub. These MPAs are likely exposed to significant anthropogenic pressures, such as urbanization, coastal development, light pollution, and tourism-related activities. Their proximity suggests that these MPAs may face increased risks related to human activities, such as water pollution, habitat degradation, and disturbances to marine life. The closeness of these MPAs to an urban center implies the need for focused management strategies that mitigate urban impacts, promote sustainable tourism, enforce regulations, and enhance community-led conservation efforts to preserve these critical marine areas amidst increasing human pressure.

While this might be the case for cluster 5 MPAs, majority of MPAs are concentrated around Cebu and Bohol. Most of these MPAs still have diverse marine biodiversity and most of them are only slightly impacted by light pollution from urbanization. This suggests that, despite their proximity to urban centers like Cebu City and other populated areas in Bohol, many of these MPAs have managed to retain their diverse marine biodiversity. The lower impact of urbanization and light pollution on these MPAs could indicate the presence of successful local conservation programs, community engagement in marine protection, and potentially effective enforcement of environmental regulations. Additionally, the diverse marine ecosystems, including coral reefs, seagrass meadows, and mangrove forests, may act as natural buffers, reducing the effects of light pollution and other urban influences. These ecosystems provide important services, such as filtering pollutants and serving as barriers against anthropogenic impacts. The maintenance of biodiversity in these MPAs despite urban proximity suggests an opportunity to promote sustainable tourism, eco-friendly development, and environmentally conscious policies that balance economic activity with marine conservation. Continuous monitoring is crucial to ensure that these MPAs do not become more severely impacted over time. Adaptation strategies may include mitigating light pollution, reducing coastal development pressure, and enhancing marine habitat resilience to ensure the long-term health of these ecosystems. Meanwhile, remote areas fall within clusters 0 and 1, indicating that these MPAs are well protected from light pollution and thus retain marine biodiversity.

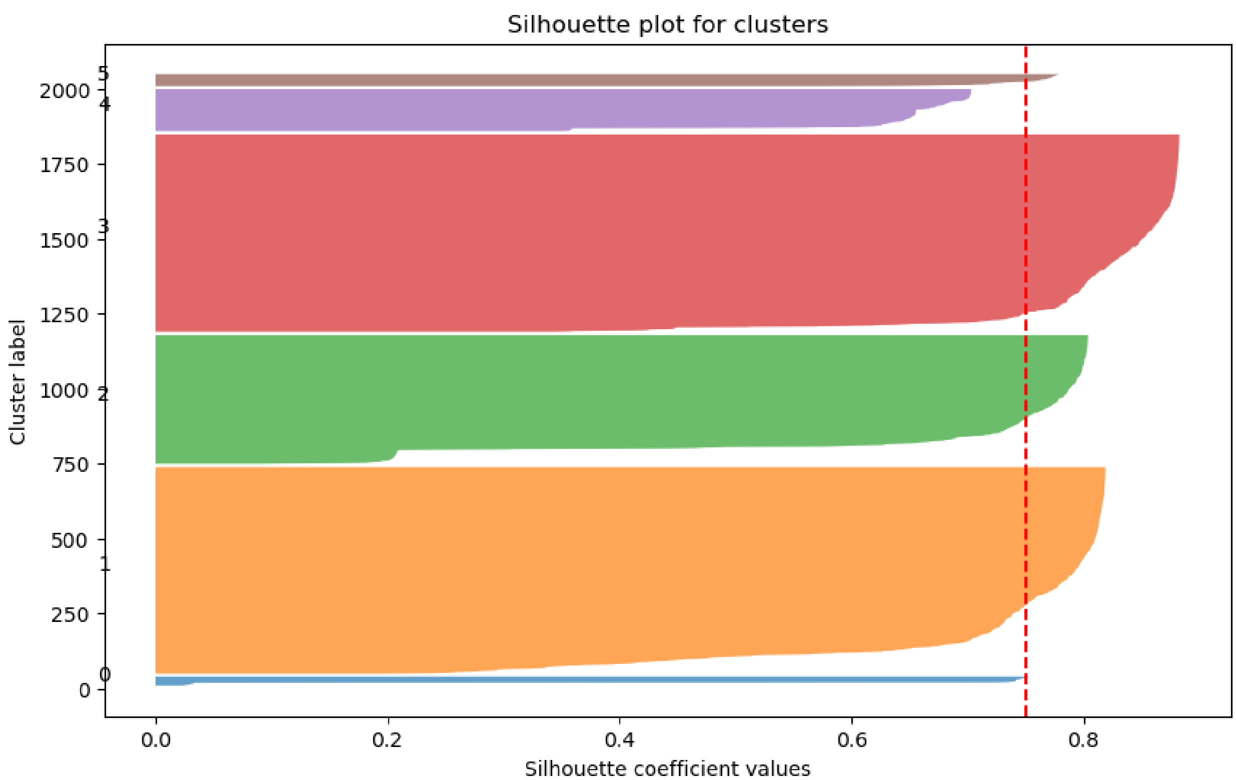

Figure 8. Silhouette Scores of Clusters.

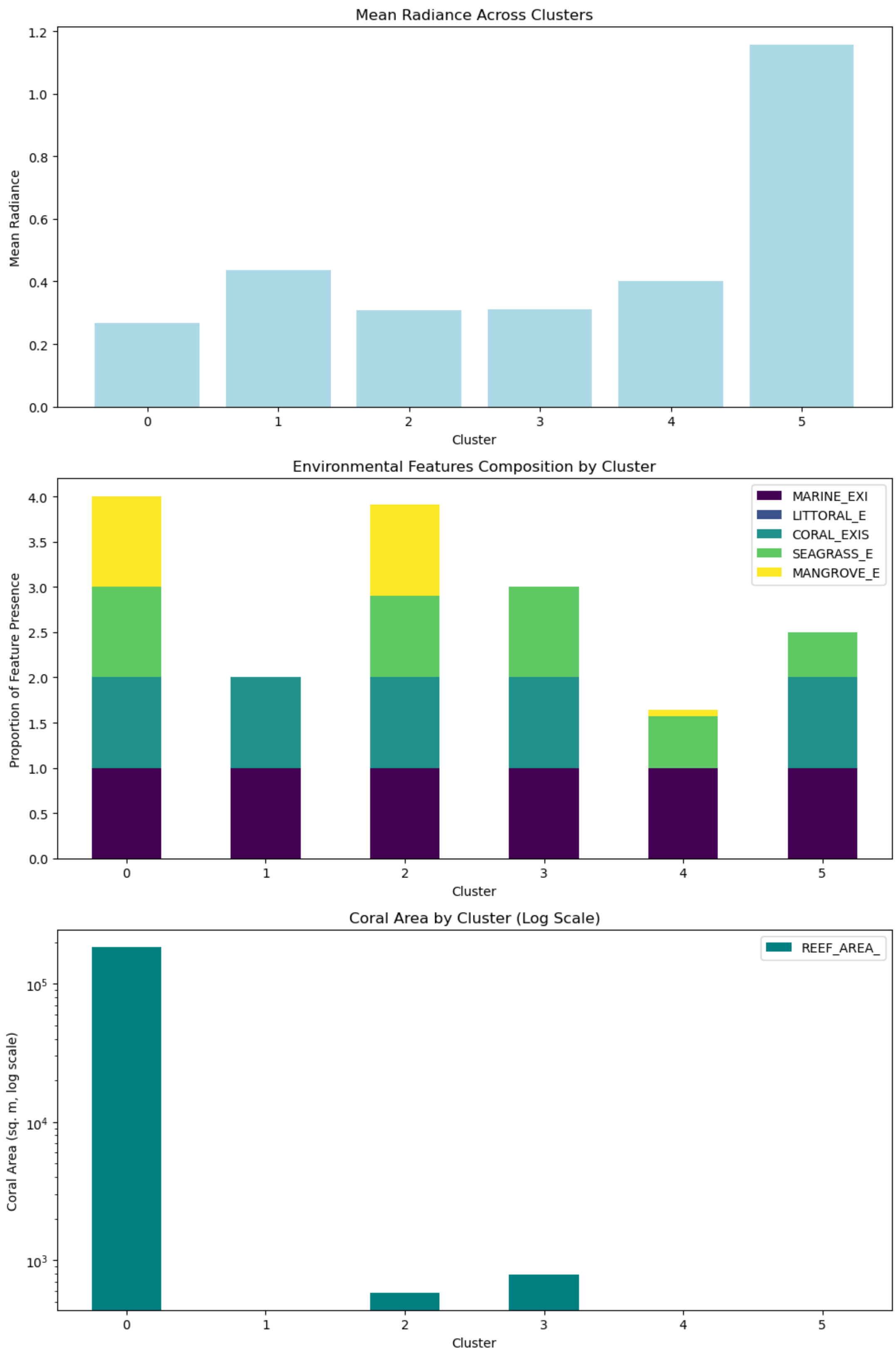

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Figure 9. Characteristics of Each Cluster.

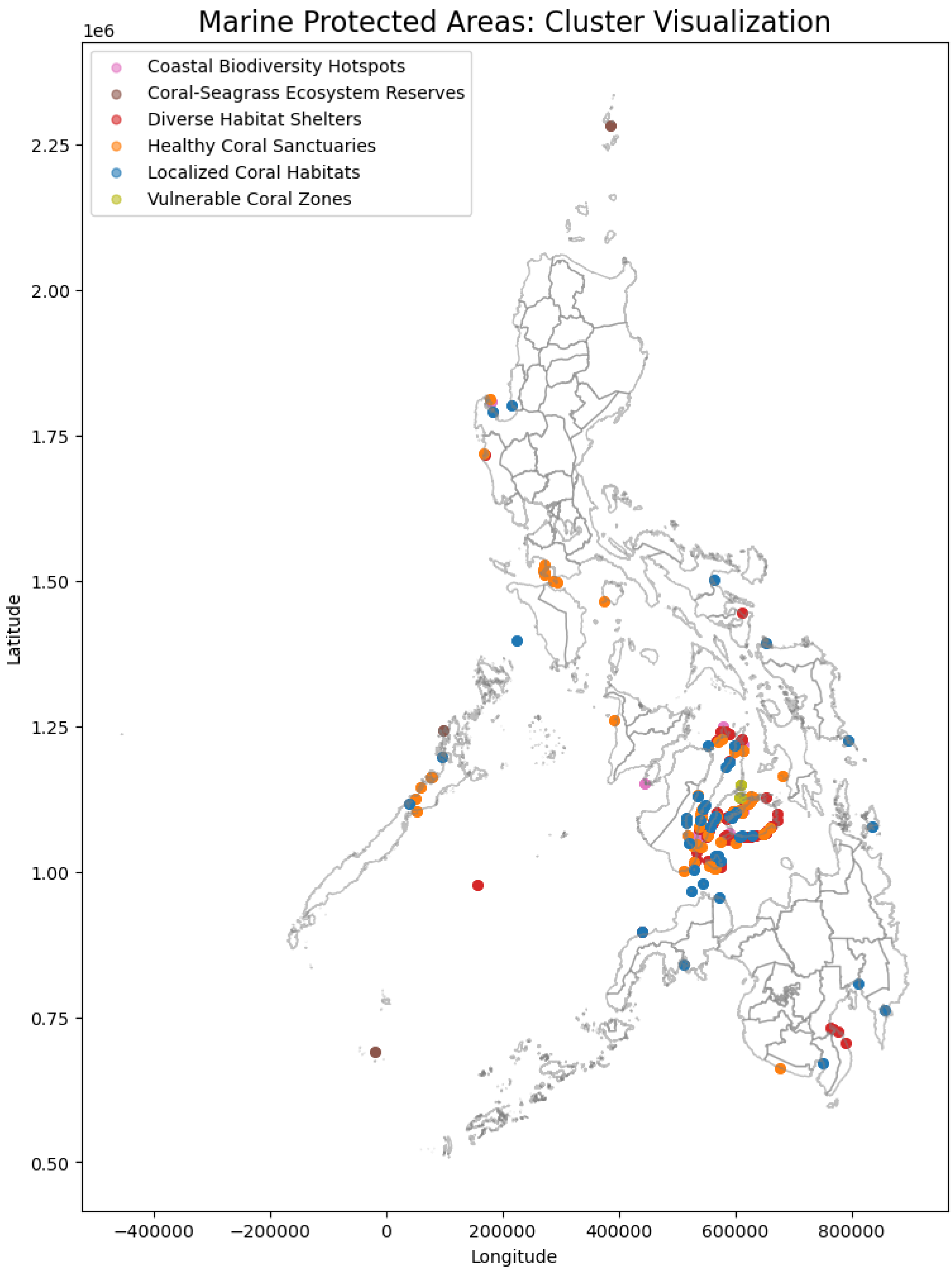

Figure 10. Locations of MPAs by Cluster.

Using this method of grouping MPAs, appropriate actions can be taken to preserve marine life and manage these habitats in a sustainable manner. Cluster 5 MPAs, being the most impacted by light pollution, must be immediately shielded against it. Urban planning must also consider these marine environments. Efforts from the local communities could also be helpful in conservation and educational campaigns about the importance of these marine ecosystems and the impact of urban activities on them. Additionally, sustainable tourism programs should be implemented to minimize the impact of anthropogenic activities on the MPAs. Clusters 0 and 2, being the most ecologically rich clusters, could benefit from holistic conservation measures that protect the entire ecosystem, including coral reefs, seagrass meadows, and mangrove forests, to maintain ecological interdependence and resilience. Since they contain abundant sea life, regular monitoring programs should be established to track changes in biodiversity, coral cover, and environmental threats, allowing for timely interventions. Cluster 4, which are mangrove zones, must be protected and restored, recognizing the role of mangrove forests as natural buffers that protect other ecosystems from anthropogenic impacts. In general, for all clusters, long-term monitoring systems should be implemented to assess environmental health, light pollution levels, and changes in marine biodiversity. There should also be educational programs in place to raise awareness about the importance of marine biodiversity and the impacts of human activity on MPAs. Furthermore, partnerships with local communities, NGOs, government agencies, and international organizations would be beneficial to support conservation and sustainable management efforts across all MPAs.

Future risk scores were predicted using two models: ARIMA and Prophet. With historical data from 2014 to 2022 as the training set, the predictions were verified using the 2023 data. Figures 11 and 12 shows the risk score in 2023 as predicted by ARIMA and Prophet plotted alongside the actual risk score in 2023. ARIMA forecast shows a decline in the average risk score from 2022 for all MPAs, while Prophet expects it to increase. The RMSE for both models are compared in Table 1. Prophet can be seen to perform significantly better than ARIMA.

Table 1. Root mean squared error for 2023 predictions for both models.

Model	RMSE
ARIMA	0.2312
Prophet	0.1579

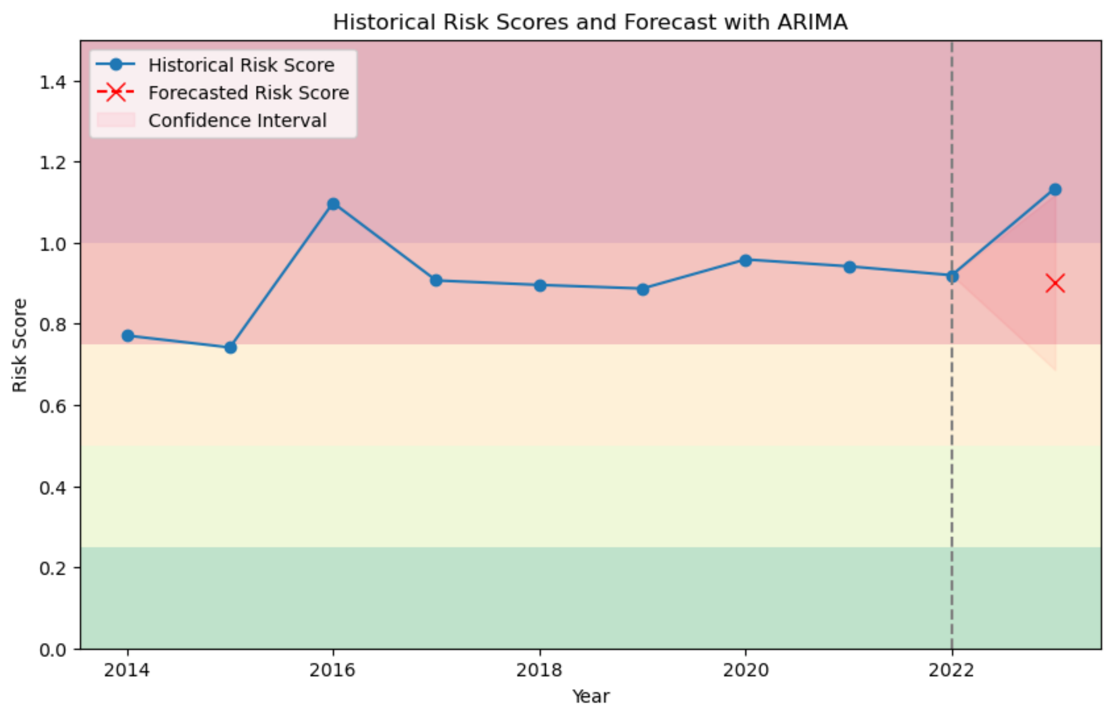

Figure 11. Risk Score Forecast for All MPAs with ARIMA.

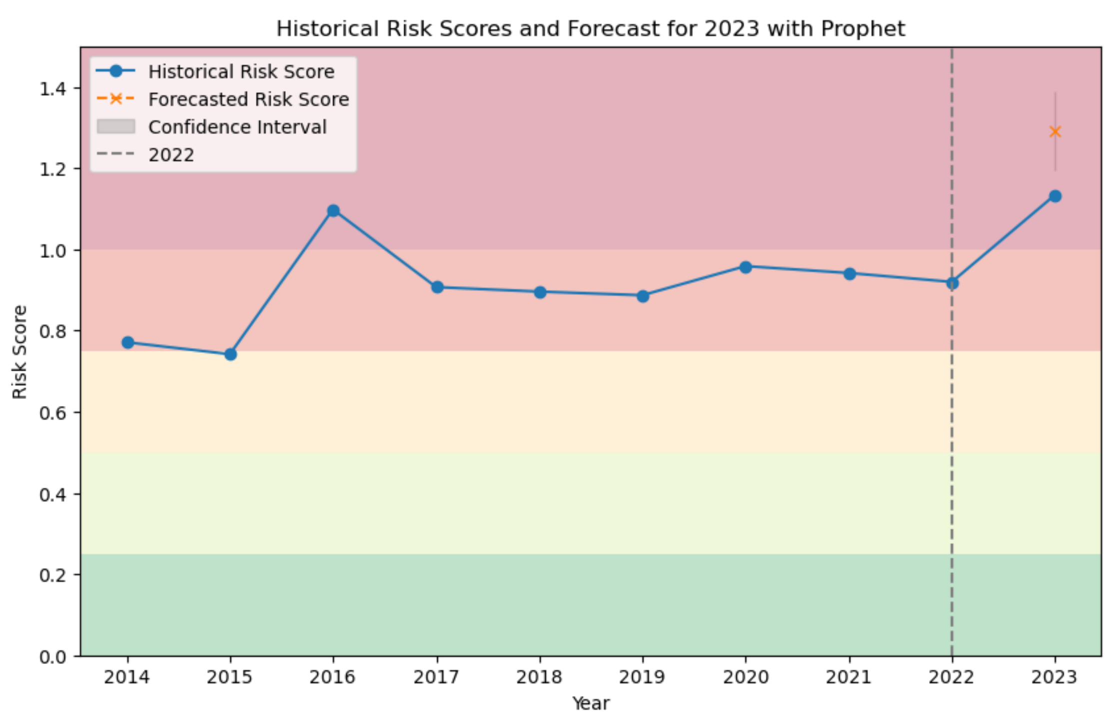

Figure 12. Risk Score Forecast for All MPAs with Prophet.

The risk scores were also predicted per cluster, shown in Figures 13 and 14. The performances of the models per cluster are summarized in Table 2. This table shows that ARIMA performs slightly better than Prophet. However, taking into consideration the generalized model for all MPAs, Prophet performs better in general. Looking at the RMSE for each cluster, Prophet performs better for most of the clusters except cluster 0 and 5. Cluster 5 could be considered an outlier since it varies greatly from the other data points. Therefore, the Prophet should be used in predicting future risk scores.

Table 2. Comparison of root mean squared error for each model per cluster.

Cluster	**RMSE**	
	ARIMA	**Prophet**
0	0.1206	0.1394
1	0.2142	0.1743
2	0.1981	0.1394
3	0.2419	0.1242
4	0.2409	0.1749
5	0.2247	0.5073
Mean RMSE	*0.2067*	*0.2099*

Finally, this model is used to predict future risk scores for the next five years. This is important for decision makers to identify which clusters to prioritize in conservation efforts. The predictions are plotted with historical data in Figure 15. Cluster 5, being the cluster with historically highest risk scores, is projected to have even higher risk score in the future. However, a wide confidence interval indicates that higher uncertainty in future predictions. This trend underscores the intense anthropogenic pressures, such as urbanization and light pollution, likely facing Cluster 5 MPAs, highlighting a critical need for targeted interventions. Meanwhile, other clusters exhibit lower and relatively stable historical risk scores compared to Cluster 5. Their risk trends over time show limited increases, and the forecasted values suggest that risks may remain stable or experience moderate increases over the next five years. This stability indicates a more resilient state of these MPAs to existing threats, potentially due to effective conservation measures or lower human impact compared to Cluster 5. These clusters offer an opportunity to reinforce existing measures, ensuring long-term resilience and sustainability. Continuous monitoring and adaptable conservation strategies will be essential across all clusters to address future risks effectively.

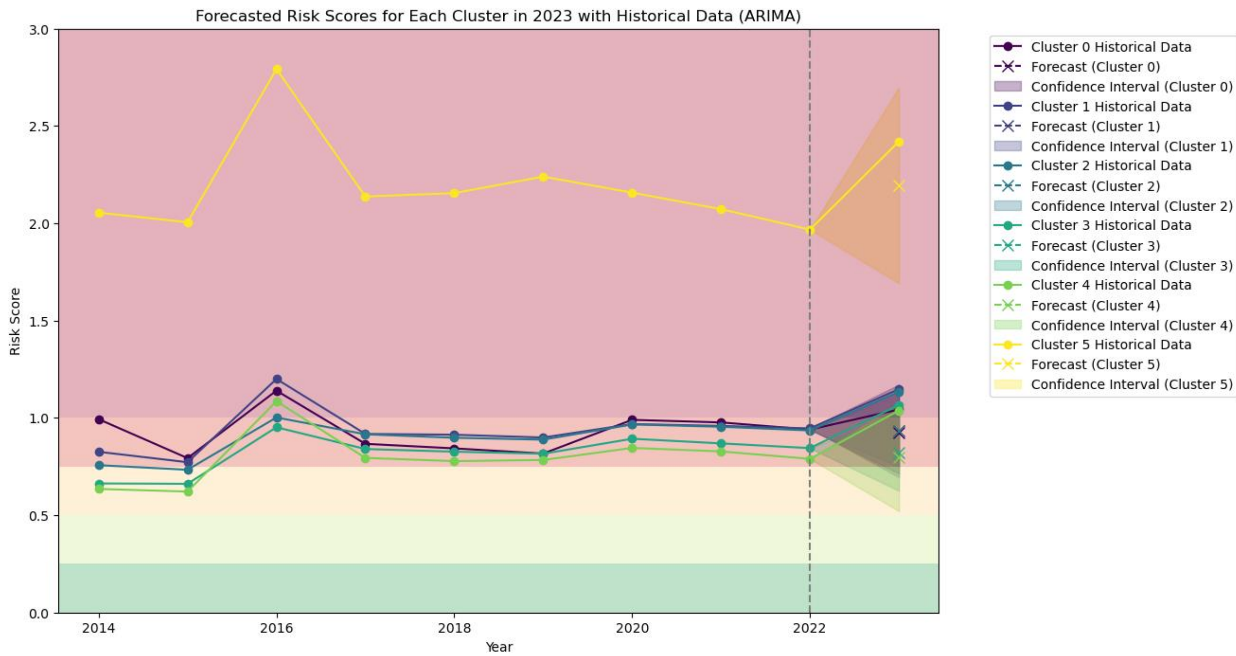

Figure 13. Risk Score Forecasts for MPAs per Cluster with ARIMA.

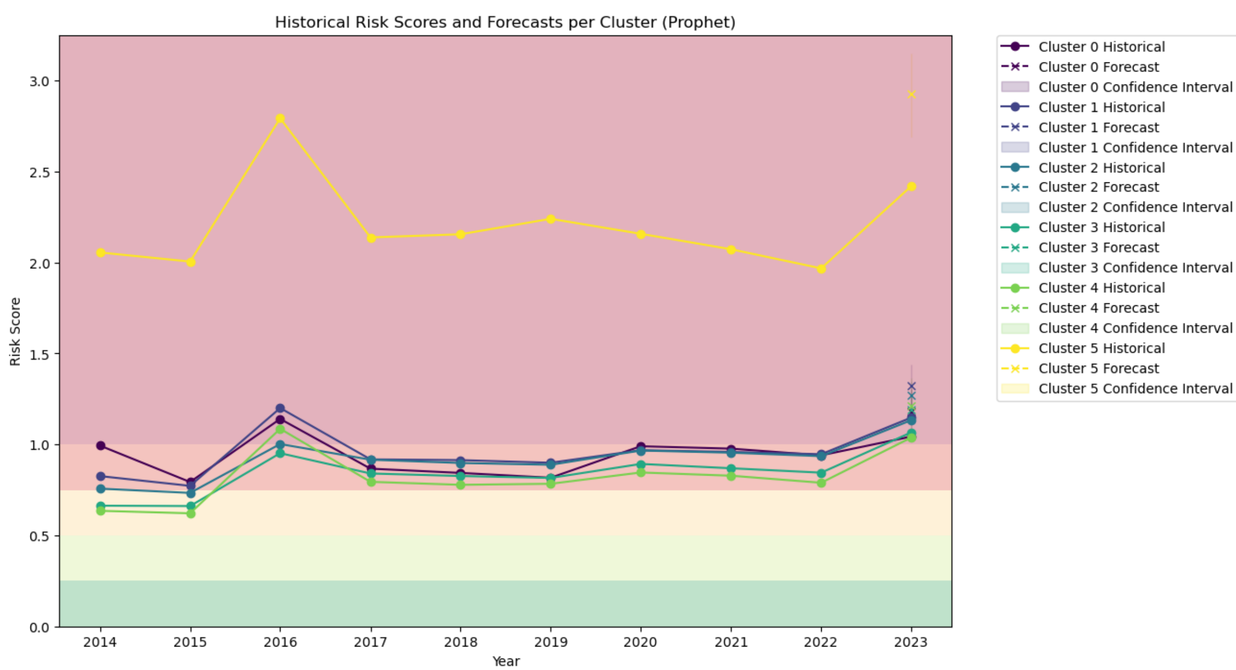

Figure 14. Risk Score Forecasts for MPAs per Cluster with Prophet.

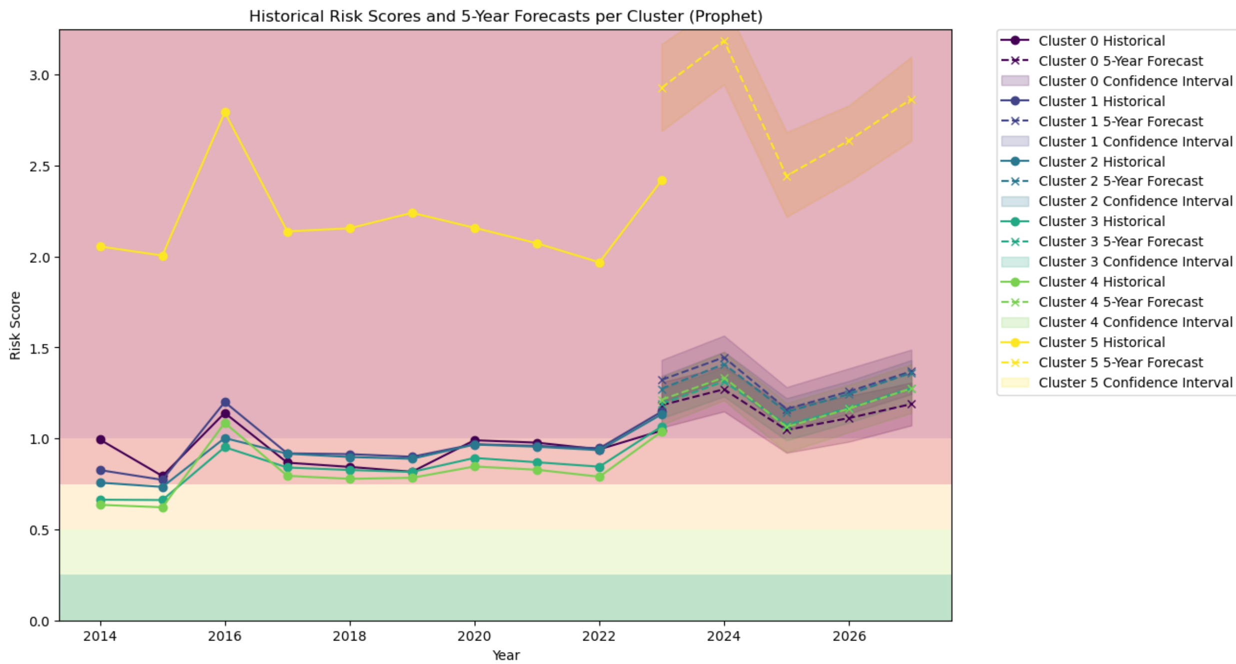

Figure 15. 5-year Risk Score Forecasts for MPAs per Cluster with Prophet.

V. Conclusion

Results of this study show that light pollution in MPAs in the Philippines has been rising over time, endangering marine biodiversity. Clustering of MPAs, which shows different groups of MPAs based on ecological factors and exposure to light pollution, show that MPAs closest to metropolitan cities such as Cebu City have the greatest radiance. On the other hand, other clusters classified as ecologically rich, which are home to a variety of ecosystems such as mangroves, seagrass meadows, and coral reefs, show comparatively lower levels of light pollution. They are resilient, but they might potentially become vulnerable if left unguarded. Light pollution hazards are expected to continue to rise, particularly for MPAs receiving significant amounts of light pollution, according to risk score projections utilizing Prophet and ARIMA models. These results demonstrate the urgent need for focused conservation efforts that are adapted to the particular traits of every cluster. While maintaining the resilience of those less affected, strategies must give priority to the MPAs that are more at risk. It is impossible to overestimate the significance of adaptable urban development, sustainable tourism, and community involvement in reducing human stresses on these marine ecosystems.

VI. Recommendations

The findings in this study suggest that protect Marine Protected Areas in the Philippines from light pollution, there should be a collaboration between the government, community, non-governmental organizations, and foreign partners. Efforts should be made to overall mitigate light exposure on MPAs, prioritizing cluster 5 MPAs. Measures like shielded lighting, curfews on lighting use, and the encouragement of "dark sky" projects to reduce artificial light pollution in coastal regions should be implemented. Local government entities should integrate the requirements of marine conservation into plans for urban growth, with a focus on environmentally friendly methods that reduce pollution from light and other sources. Local communities could get involved in promoting awareness about marine habitats and how to conserve them. Ecologically rich MPAs should be protected and if necessary, restored, so that the marine ecosystem of the Philippines would not be threatened, since these MPAs are the breeding grounds of marine life. MPAs will benefit from long-term monitoring programs for marine biodiversity and light pollution to enable prompt responses and flexible management techniques as risks change. Since tourism is one of the major causes of light pollution, policies and procedures for eco-friendly travel that balance financial gains with environmental protection should be implemented. Along with policies, laws that protect MPAs from light pollution should be strengthened and supported with strong enforcement measures.

VII. References

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